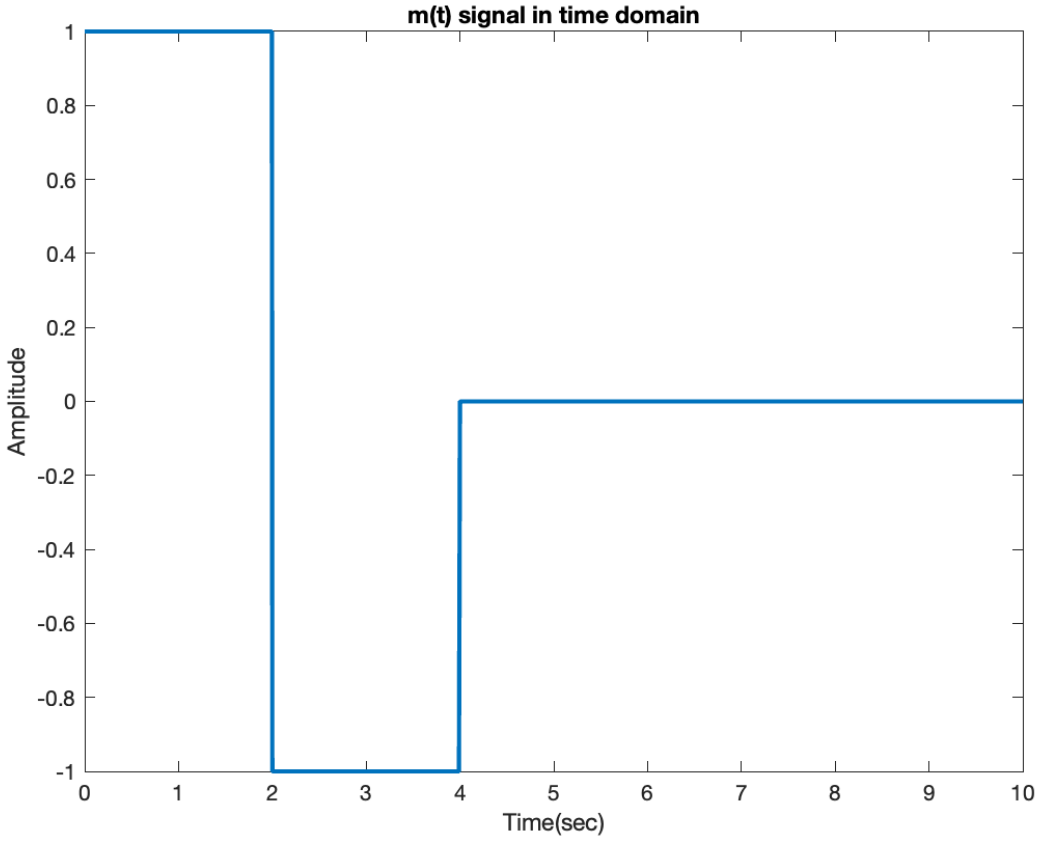
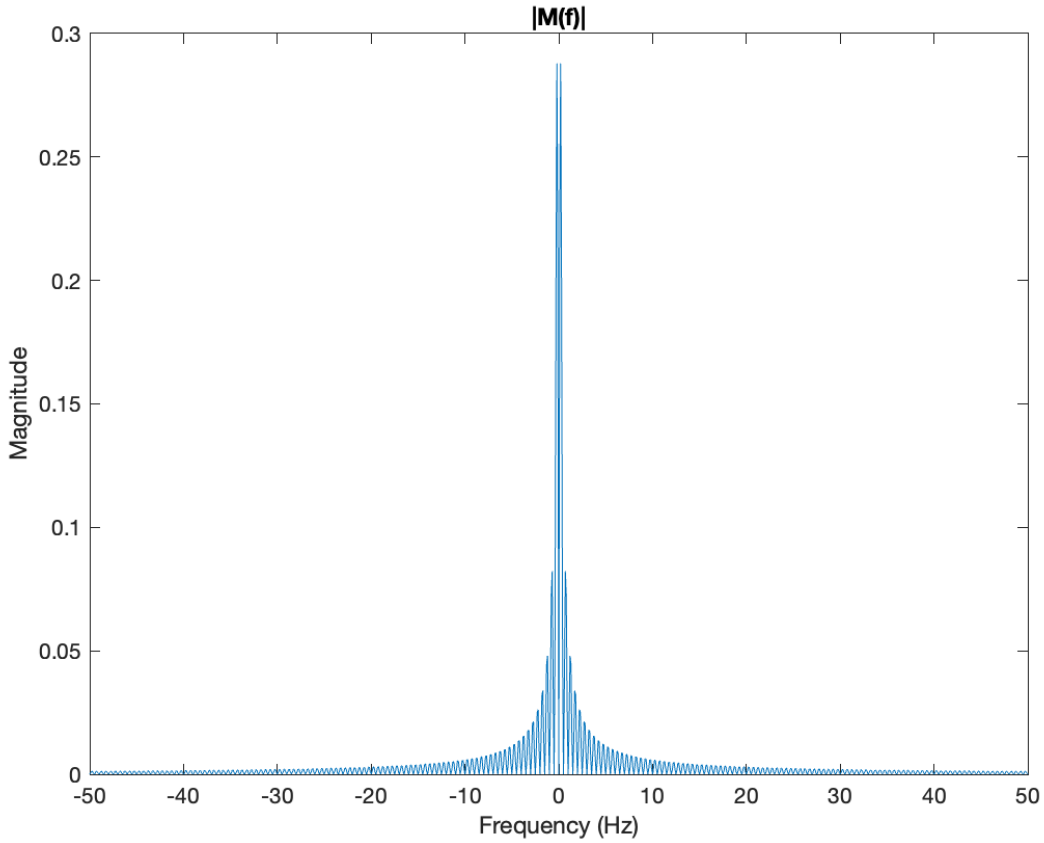
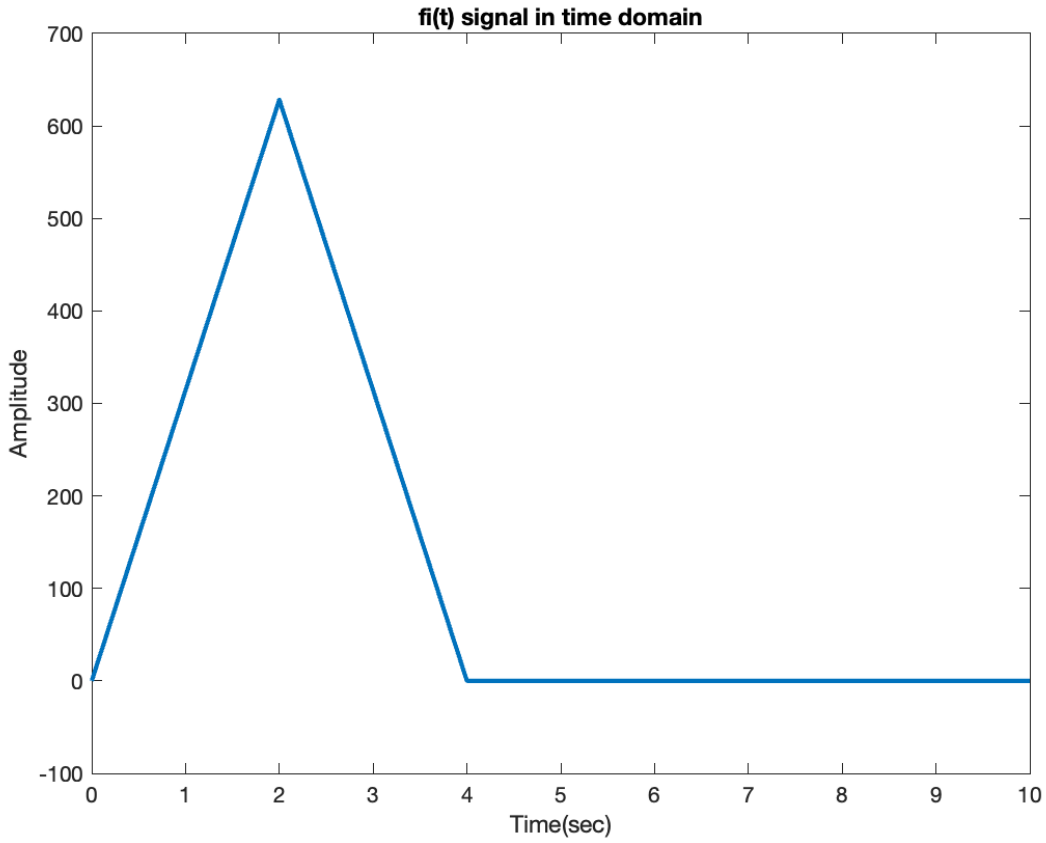
1 Analytic solutions

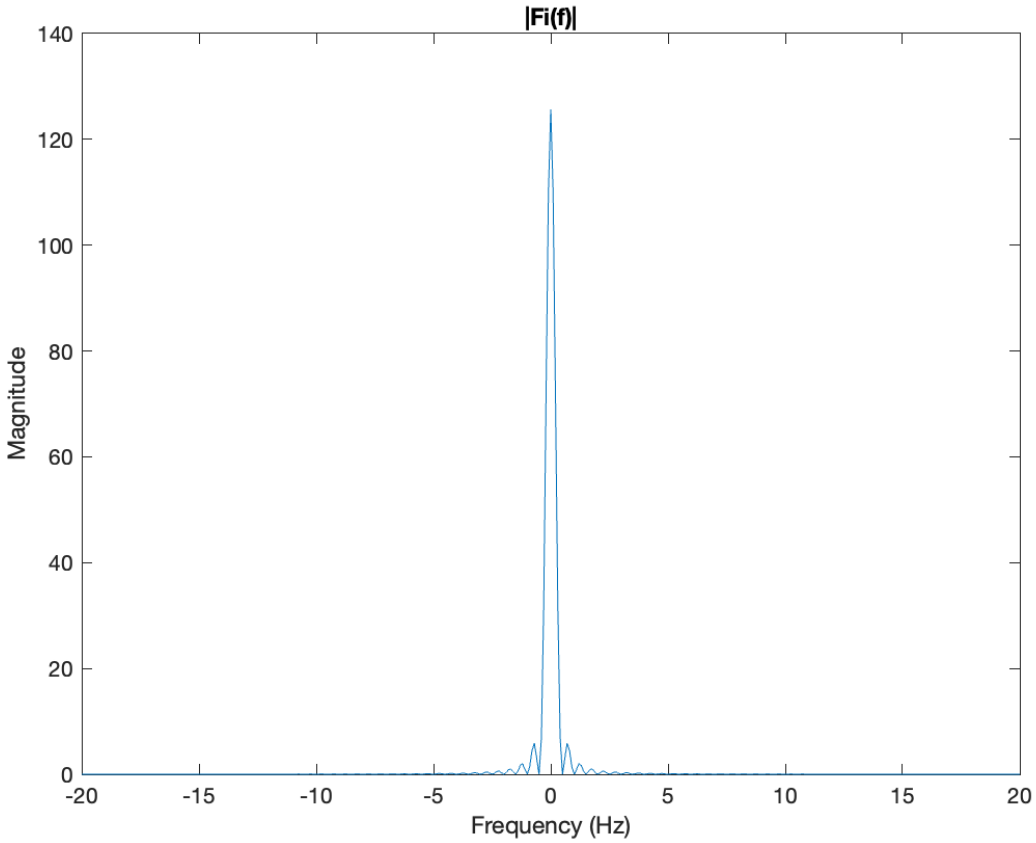
2 MATLAB solutions

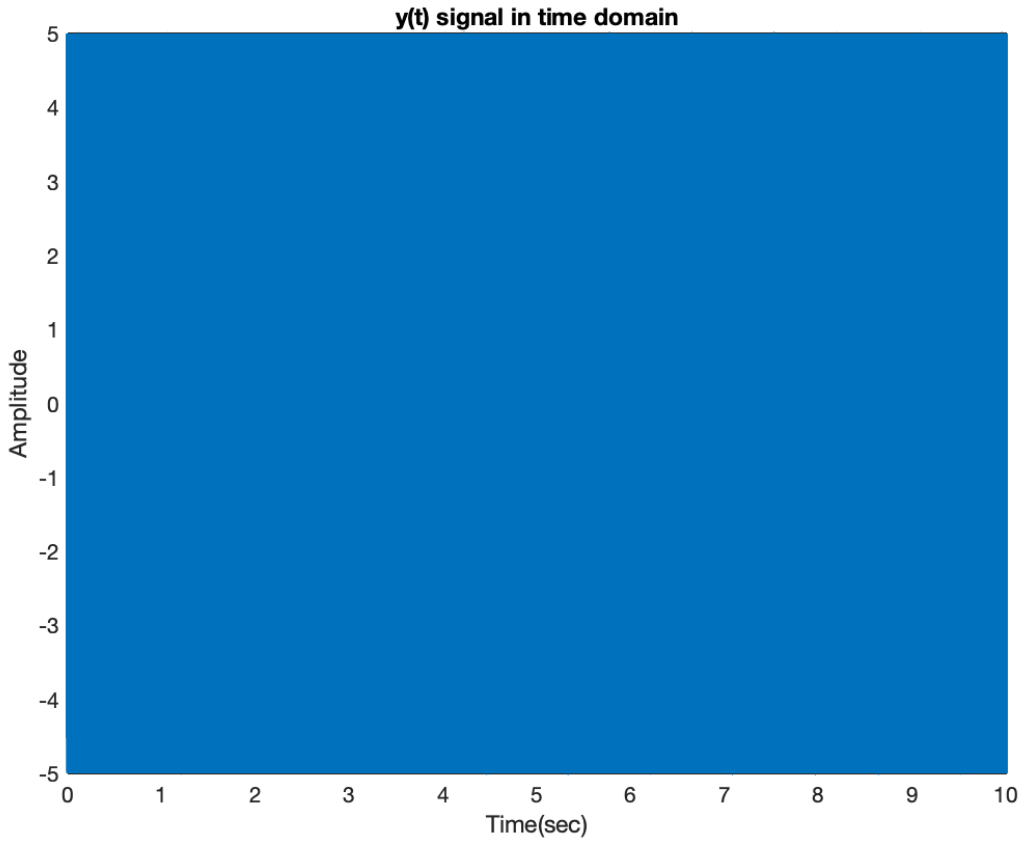
2.1 Plot of m(t) signal. M(f) signal and the plot of the magnitude spectrum

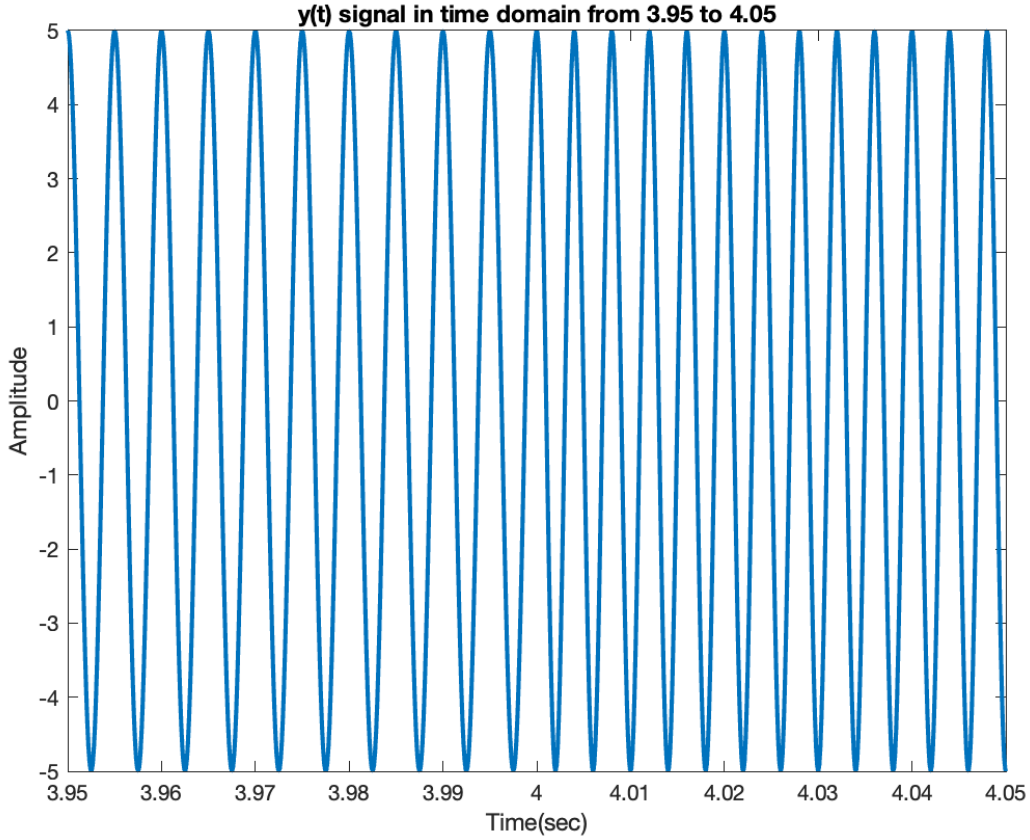

m(t) signal in time domain
Amplitude
Time(sec)
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
1
2
3
4
5
6
7
8
9
10

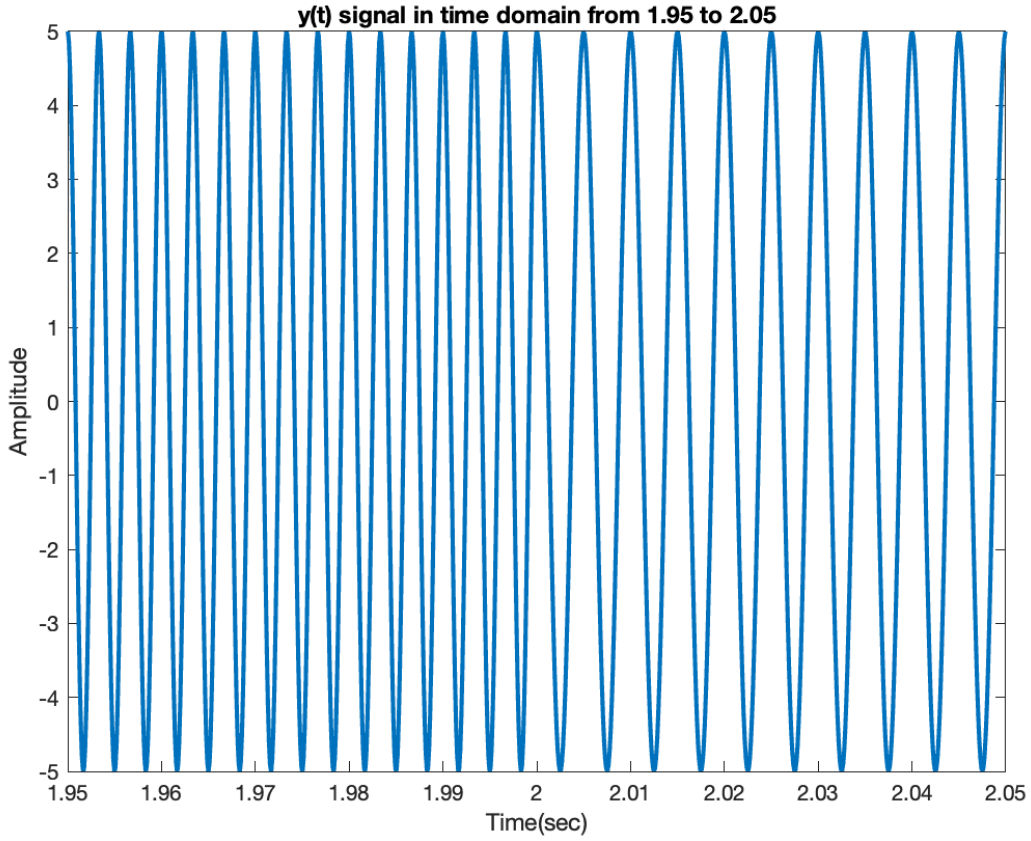

|M(f)|
Magnitude
Frequency (Hz)
0.3
0.25
0.2
0.15
0.1
0.05
0
-50
-40
-30
-20
-10
0
10
20
30
40
50

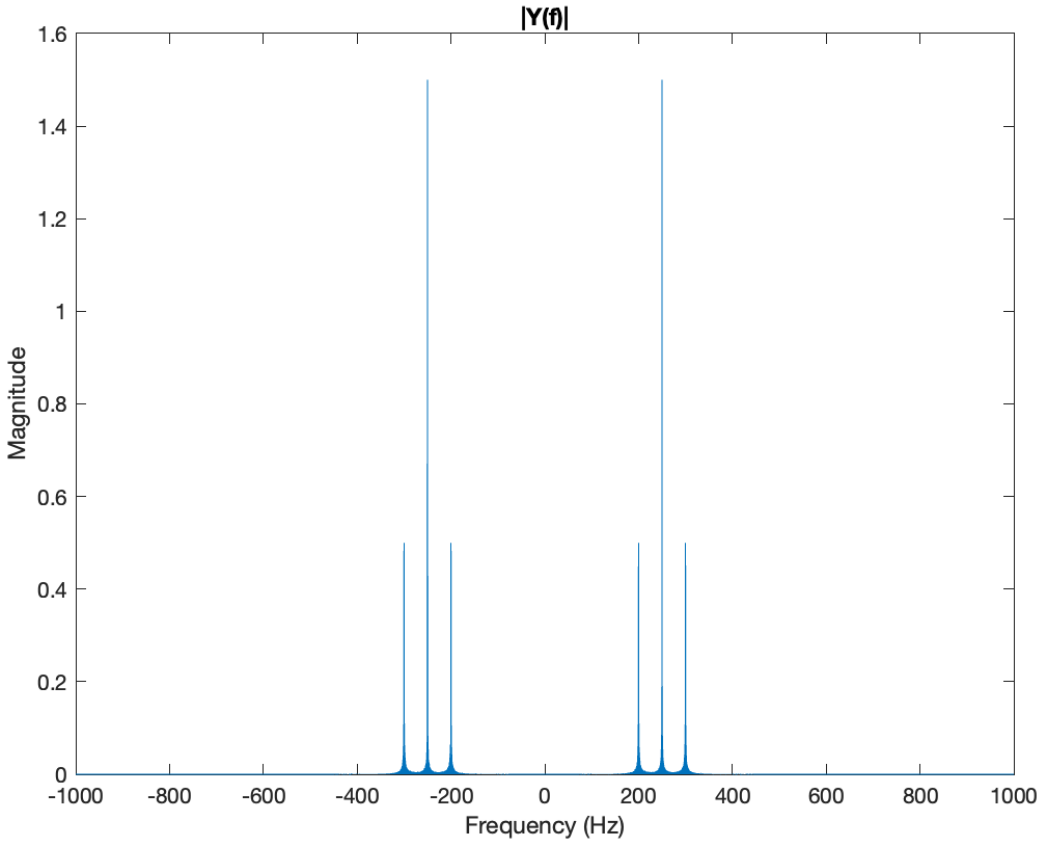

fi(t) signal in time domain
Amplitude
Time(sec)
700
600
500
400
300
200
100
0
-100
0
1
2
3
4
5
6
7
8
9
10

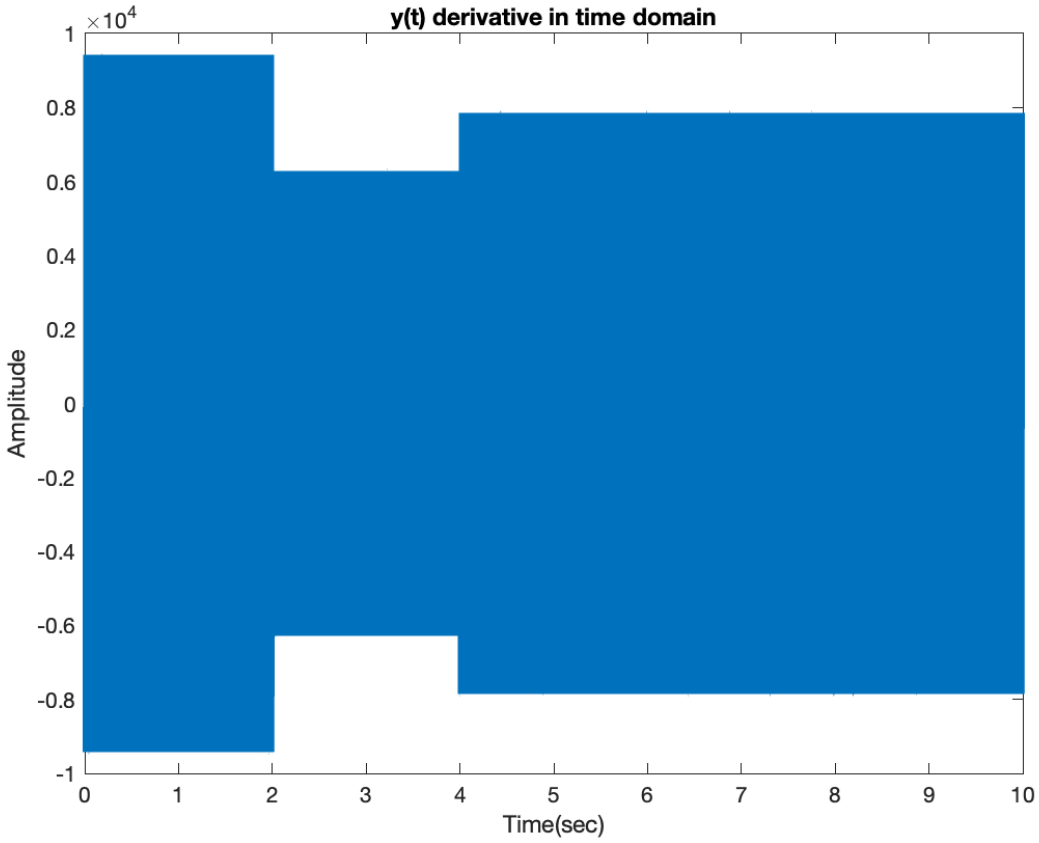

|Fi(f)|
Magnitude
Frequency (Hz)
140
120
100
80
60
40
20
0
-20
-15
-10
-5
0
5
10
15
20

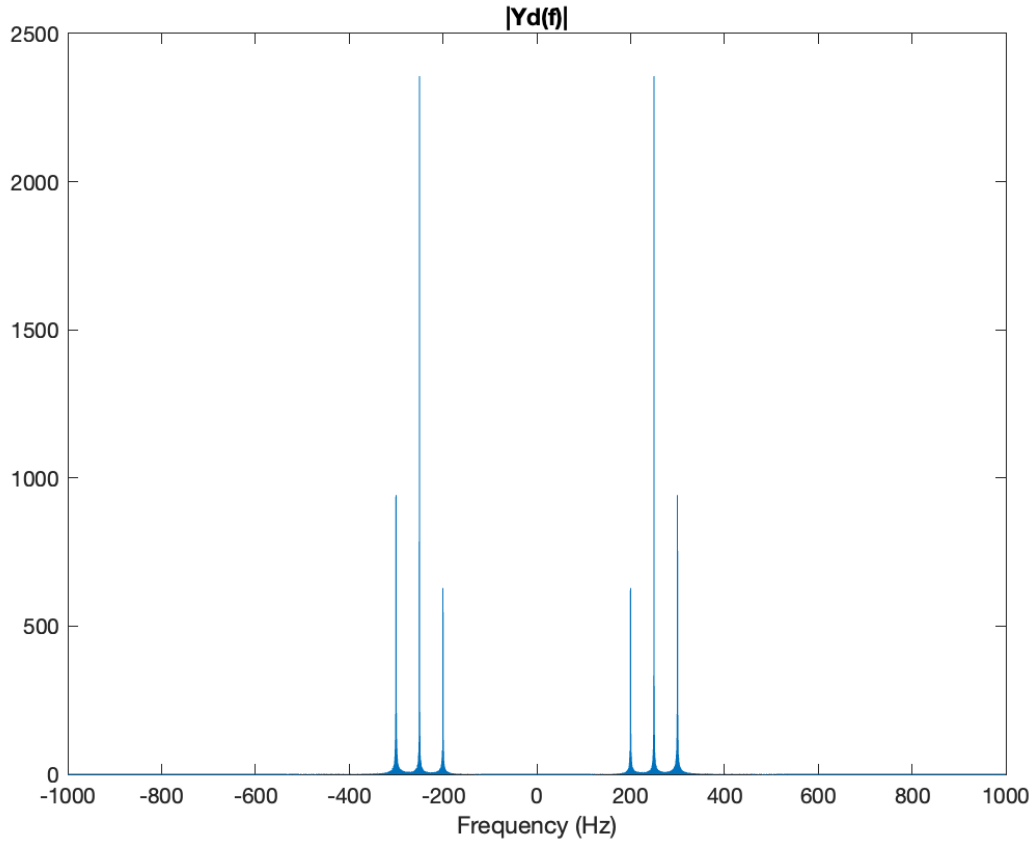

y(t) signal in time domain
Amplitude
Time(sec)
5
4
3
2
1
0
-1
-2
-3
-4
-5
0
1
2
3
4
5
6
7
8
9
10

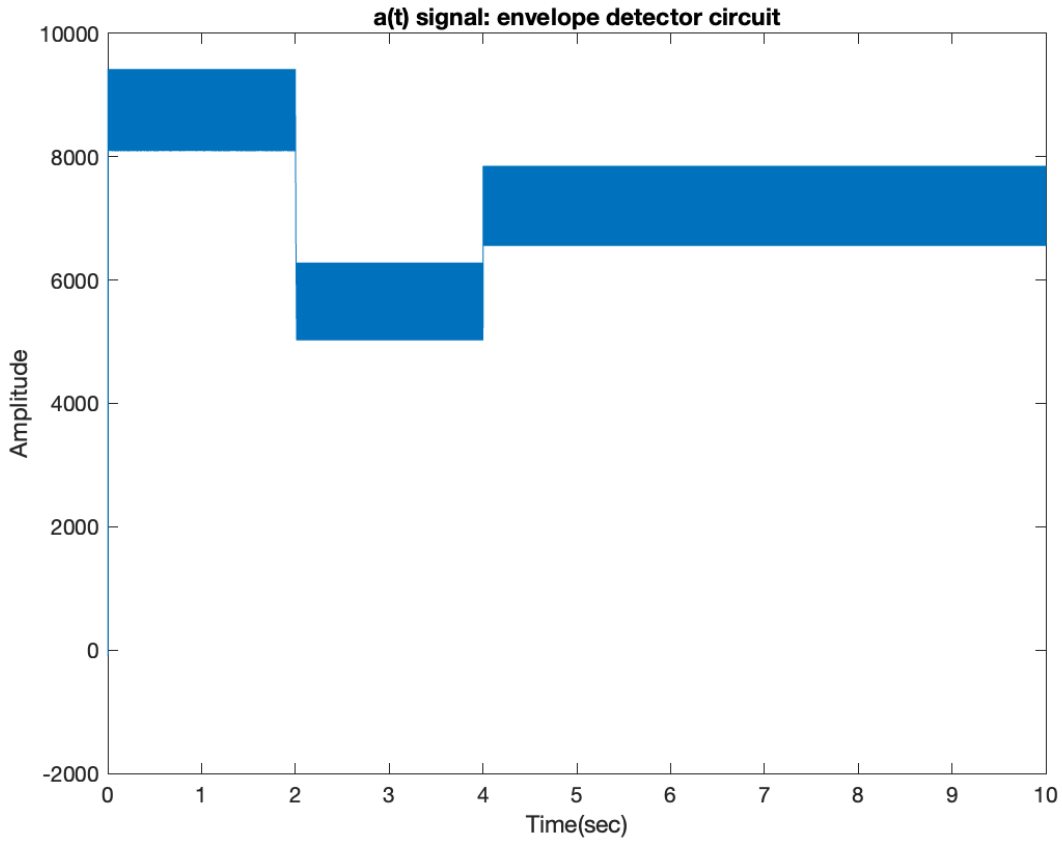

y(t) signal in time domain from 3.95 to 4.05
Amplitude
Time(sec)
5
4
3
2
1
0
-1
-2
-3
-4
-5
3.95
3.96
3.97
3.98
3.99
4
4.01
4.02
4.03
4.04
4.05

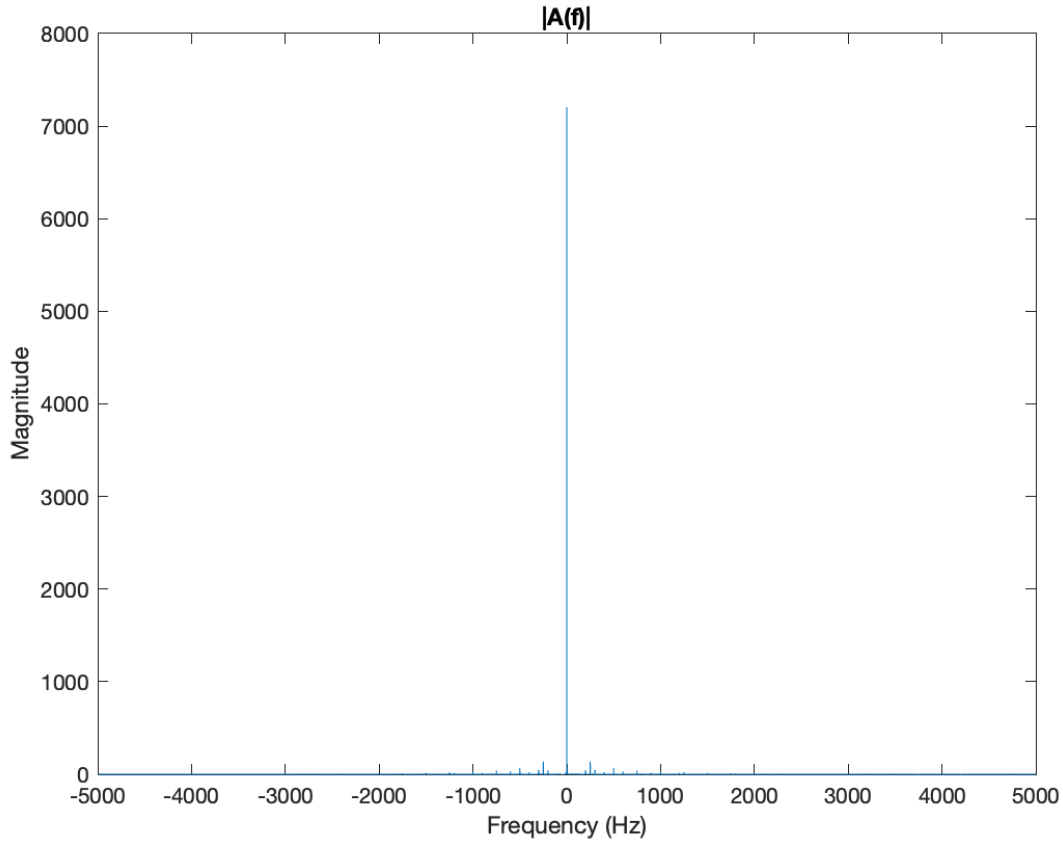

y(t) signal in time domain from 1.95 to 2.05
Amplitude
Time(sec)
5
4
3
2
1
0
-1
-2
-3
-4
-5
1.95
1.96
1.97
1.98
1.99
2
2.01
2.02
2.03
2.04
2.05

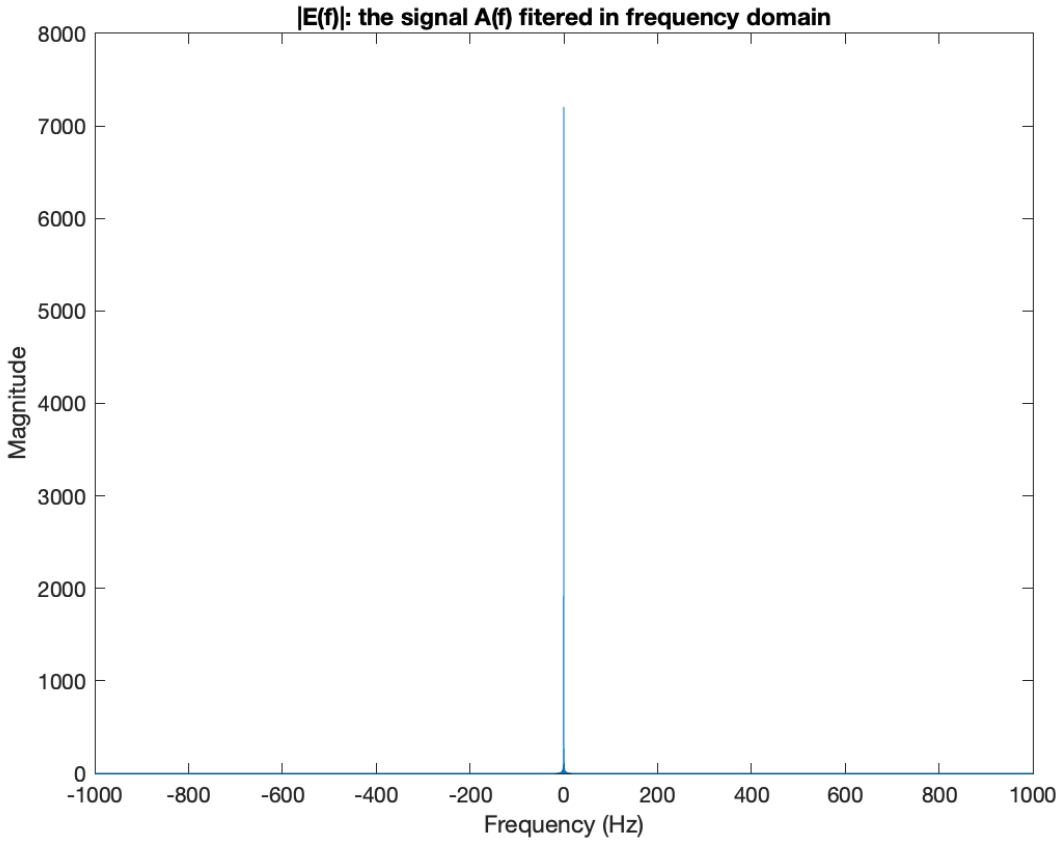

|Y(f)|
Magnitude
Frequency (Hz)
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
-1000
-800
-600
-400
-200
0
200
400
600
800
1000

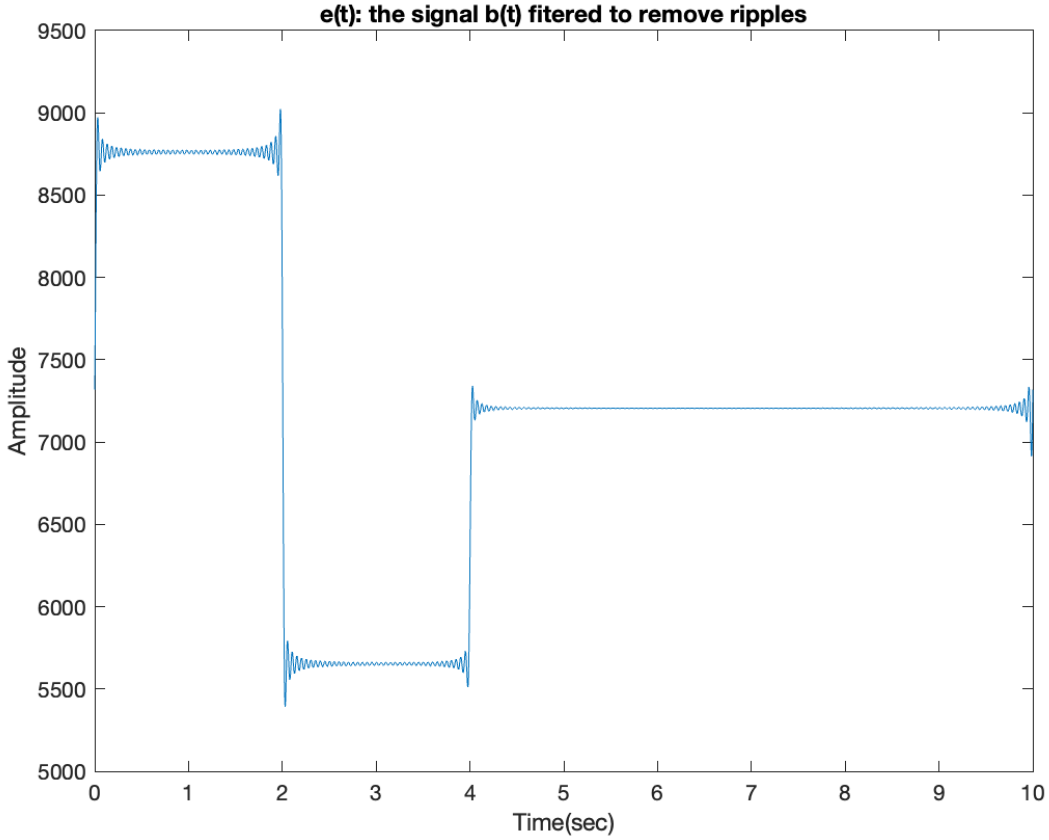

y(t) derivative in time domain
×10^4
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
Amplitude
0
1
2
3
4
5
6
7
8
9
10
Time(sec)

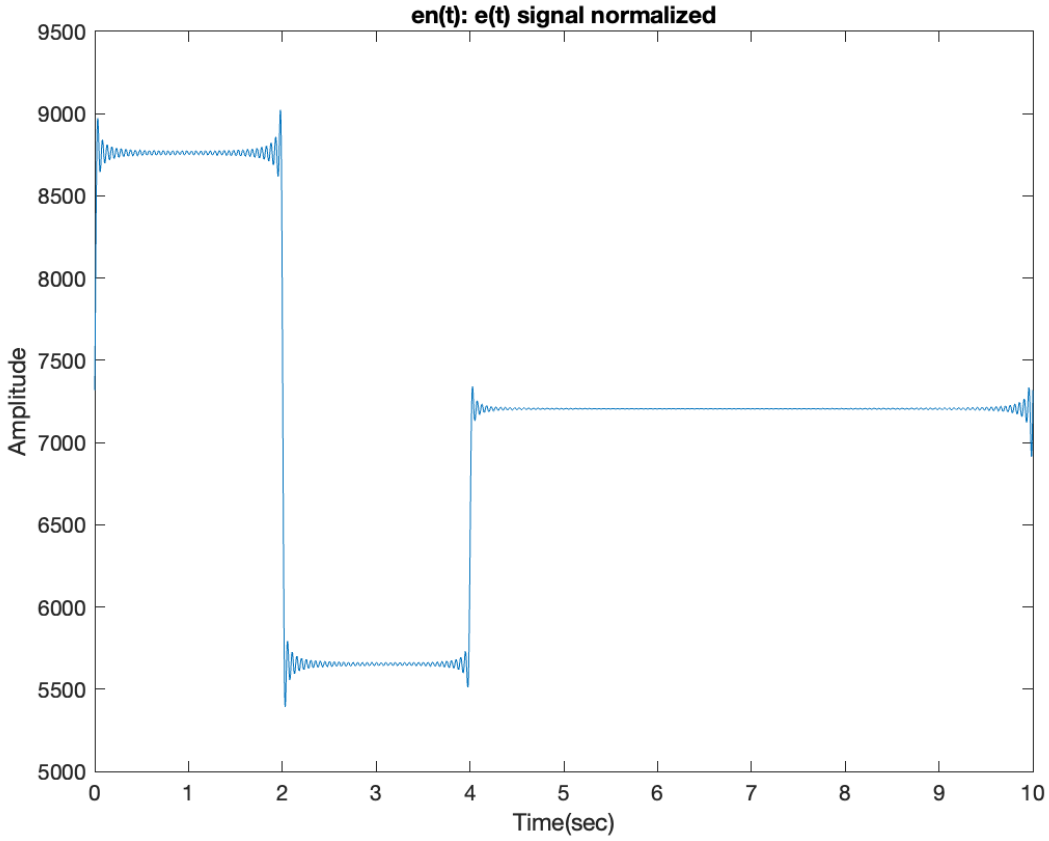

|Yd(f)|
2500
2000
1500
1000
500
0
-1000
-800
-600
-400
-200
0
200
400
600
800
1000
Frequency (Hz)

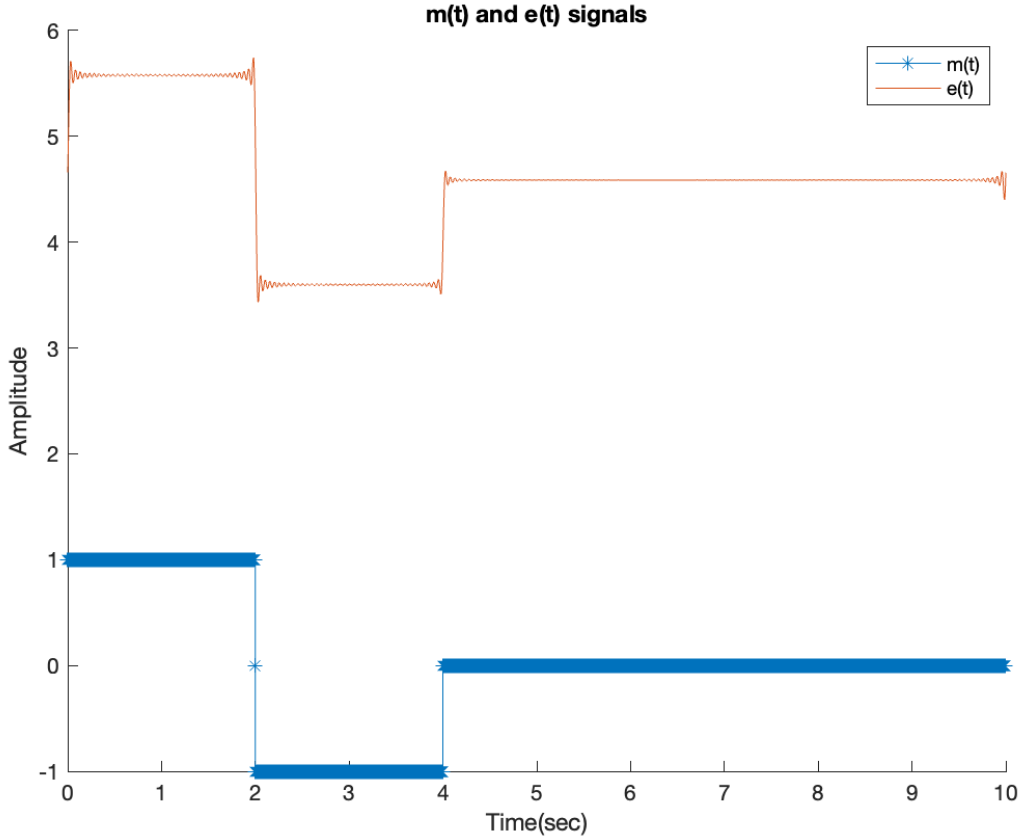

a(t) signal: envelope detector circuit
10000
8000
6000
4000
2000
0
-2000
Amplitude
0
1
2
3
4
5
6
7
8
9
10
Time(sec)

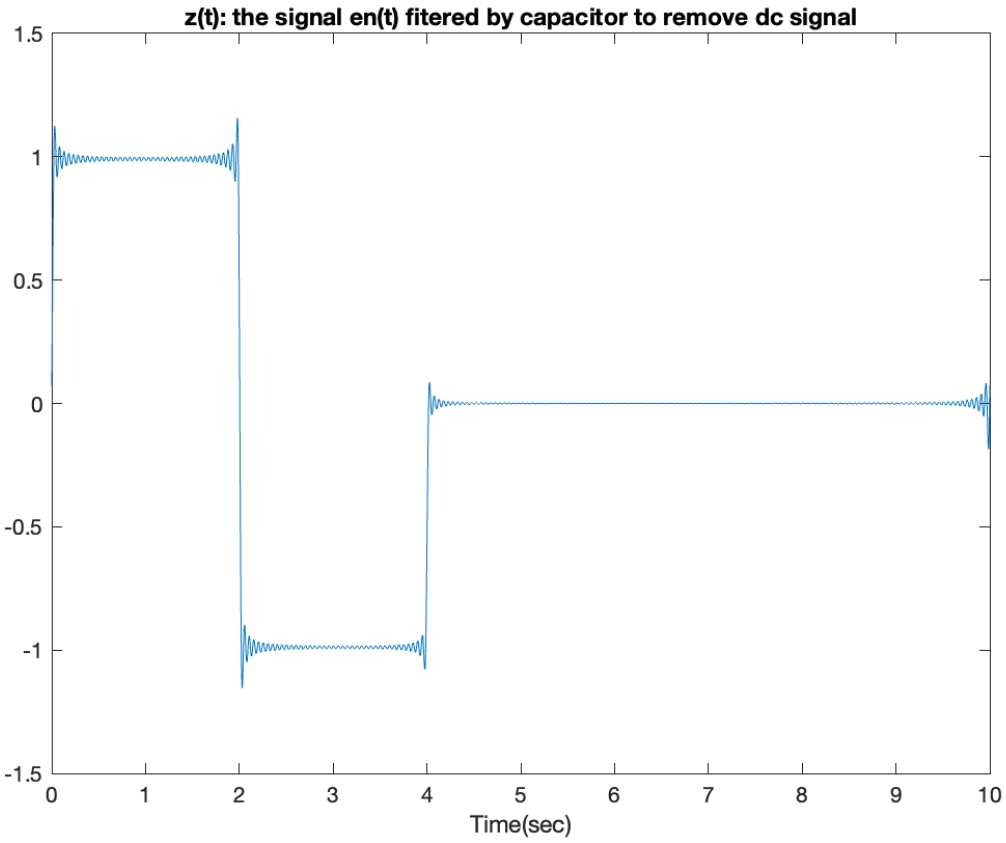

|A(f)|
8000
7000
6000
5000
4000
3000
2000
1000
0
Magnitude
-5000
-4000
-3000
-2000
-1000
0
1000
2000
3000
4000
5000
Frequency (Hz)

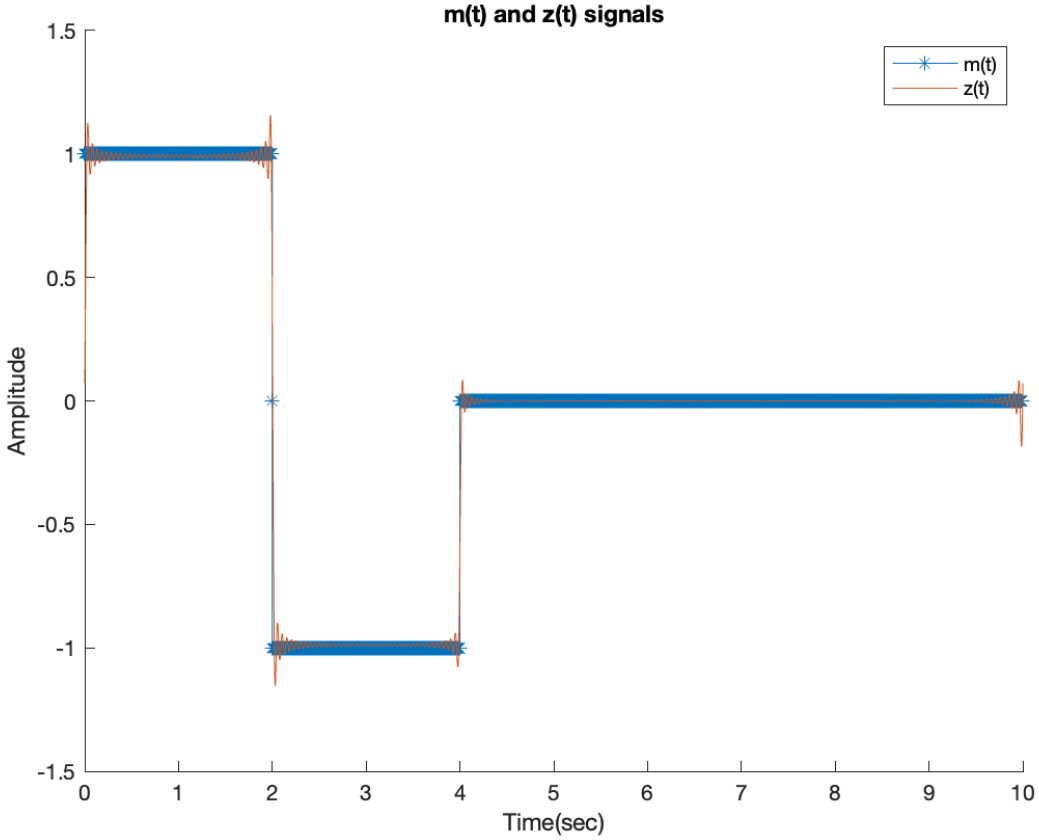

|E(f)|: the signal A(f) fitered in frequency domain
Magnitude
Frequency (Hz)
8000
7000
6000
5000
4000
3000
2000
1000
0
-1000
-800
-600
-400
-200
0
200
400
600
800
1000


e(t): the signal b(t) fitered to remove ripples
Amplitude
Time(sec)
9500
9000
8500
8000
7500
7000
6500
6000
5500
5000
0
1
2
3
4
5
6
7
8
9
10


en(t): e(t) signal normalized
Amplitude
Time(sec)
9500
9000
8500
8000
7500
7000
6500
6000
5500
5000
0
1
2
3
4
5
6
7
8
9
10


m(t) and e(t) signals
m(t)
e(t)
Amplitude
Time(sec)
6
5
4
3
2
1
0
-1
0
1
2
3
4
5
6
7
8
9
10


z(t): the signal en(t) fitered by capacitor to remove dc signal
1.5
1
0.5
0
-0.5
-1
-1.5
0
1
2
3
4
5
6
7
8
9
10
Time(sec)


m(t) and z(t) signals
m(t)
z(t)
Amplitude
1.5
1
0.5
0
-0.5
-1
-1.5
0
1
2
3
4
5
6
7
8
9
10
Time(sec)